

Business Intelligence – Question Bank

B.E. Computer Engineering | 2019 Pattern | Semester VIII | Subject Code: 410253C

Sources: May/Jun 2023, Nov/Dec 2023, May/Jun 2024, Nov/Dec 2024, May/Jun 2025, Nov/Dec 2025

Legend: ★ = Number of times the question has appeared across the 6 exam papers (e.g., ★★★ = appeared 3 times). Questions appearing only once have no star.

UNIT 3 – BI Reporting, Data Models & Dashboards

- Q1. Explain in detail Drill Up & Drill Down. ★★★
- Q2. Explain the Multidimensional Data Model with a suitable example. ★★★★★
- Q3. What is data grouping and sorting? Write an example of each. ★★★
- Q4. Explain different types of reports in detail. ★★★
- Q5. Explain the Relational Data Model with example. ★★★
- Q6. Write short note on filtering reports. ★★
- Q7. What are the important BI reporting practices? ★★
- Q8. Discuss the importance of Drill Up, Drill-Down, and Drill-Through capabilities in report preparation.
- Q9. Explain with examples the use of Data Grouping and Sorting, Filtering in BI Reports.
- Q10. What are the advantages of the Multi-Dimensional Data Model?
- Q11. What is the importance of adding Conditional Formatting and Calculations in a report?
- Q12. How does the business report help any organization?
- Q13. What are the best practices in dashboard design?
- Q14. State the difference between Relational and Multidimensional Data Model. ★★
- Q15. What is a File Extension? Explain the structure of a CSV file.
- Q16. Discuss the Scatter chart and Combination chart with diagram used for reports.
- Q17. What is data integration? Explain the two methods used for the same.
- Q18. What is binning? How is it used for report creation?
- Q19. Explain Business Intelligence architectures with a suitable diagram.
- Q20. Explain the role of mathematical models in BI.
- Q21. What is data, information and knowledge? Explain with suitable examples.
- Q22. Explain Decision Support System (DSS) in detail.
- Q23. Write short note on Ethics and Business Intelligence.

UNIT 4 – Data Pre-processing, Transformation & Reduction

- Q1.** Explain data exploration in detail with example. ★ ★
- Q2.** Explain data transformation in detail with example. ★ ★ ★ ★
- Q3.** Explain data validation, incompleteness, noise, and inconsistency of quality of input data. ★ ★ ★
- Q4.** Explain data reduction in detail with example. ★ ★ ★
- Q5.** Difference between Univariate, Bivariate, and Multivariate analysis. ★ ★ ★
- Q6.** Write a short note on data discretization. ★ ★ ★
- Q7.** Discuss the need for data pre-processing and any 2 techniques used.
- Q8.** Explain Dimensionality Reduction and Data Compression. ★ ★
- Q9.** Define dirty data. What are the reasons for dirty data?
- Q10.** Explain the working of binning with a suitable example.
- Q11.** What is Bivariate analysis? Why is it important? Discuss the different types with example.
- Q12.** What is a Contingency Table? What is Marginal Distribution? Justify with a suitable example. ★ ★
- Q13.** Explain Data Reduction technique: Sampling, Feature Selection, Principal Component Analysis (PCA). ★ ★
- Q14.** Compute Mean, Median and Mode for grouped frequency data.
- Q15.** How are Mean, Median and Mode used during data cleaning?
- Q16.** Explain the need of data cleaning and different methods.
- Q17.** Explain the working of PCA and its applications.
- Q18.** Explain data discretization. What are the two methods?
- Q19.** Write short notes on data discretization.
- Q20.** Write short notes on Data Validation.

UNIT 5 – Data Mining: Classification, Clustering & Association

- Q1.** What is Association Rule Mining? Explain the terms: Support, Confidence, and Lift. ★ ★
- Q2.** What is the difference between Hierarchical Clustering and Partitioning method? ★ ★
- Q3.** Apply the Apriori Algorithm on the given dataset (TID: T1–T9, Items: I1–I5) with min support = 2 and min confidence = 60%. Find frequent item sets and generate association rules. ★ ★ ★
- Q4.** Explain Bayes Theorem in detail. ★ ★
- Q5.** Difference between Classification and Clustering. ★ ★ ★ ★
- Q6.** Explain Logistic Regression with example. ★ ★ ★ ★
- Q7.** What is Logistic Regression? Discuss the types of Logistic Regression. ★ ★
- Q8.** What is K-Means Clustering? Explain the step-by-step working of the K-Means algorithm with example. ★ ★

- Q9.** What is the Apriori Algorithm? Discuss the applications of this algorithm with example. ★ ★
- Q10.** Define: Frequent Item Set, Minimum Support Count, Hierarchical Clustering, Regression. ★ ★
- Q11.** What is a Decision Tree? Explain with a case study. ★ ★
- Q12.** What are Association Rules? How to evaluate them using Support and Confidence? Explain with example. ★ ★
- Q13.** State different formulae for Evaluation of Classification models.
- Q14.** Find clusters for the given website visitor age dataset using K-Means with $K = 2$: {16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66}.
- Q15.** Explain the advantages and disadvantages of Naïve Bayes Classifier.
- Q16.** Discuss Parameter Tuning and Optimization.
- Q17.** Write a short note on Classification.
- Q18.** Explain Association Rule and its types.

UNIT 6 – BI Tools, Applications & Industry Use Cases

- Q1.** Explain BI application in CRM (Customer Relationship Management). ★ ★ ★ ★
- Q2.** Explain Roles of Analytical Tools in Business Intelligence. ★ ★
- Q3.** Define Business Intelligence. List and explain any 3 tools for Business Intelligence. ★ ★
- Q4.** Explain applications of BI in Telecommunication and Banking. ★ ★
- Q5.** Explain BI application in Logistics and Production. ★ ★ ★
- Q6.** Explain Role of BI in Finance and Marketing. ★ ★
- Q7.** Explain advantages and Benefits of Business Intelligence in ERP. ★ ★ ★
- Q8.** What is the role of Analytics/Data Analytics in Business Intelligence? ★ ★ ★
- Q9.** Write short note on WEKA and Rapid Miner.
- Q10.** Write short note on KNIME and Rapid Miner.
- Q11.** Write short note on KNIME, BI and HR Management.
- Q12.** Justify BI is useful for Customer Relationship Management. ★ ★
- Q13.** State and elaborate similarities and differences in ERP and Business Intelligence.
- Q14.** Discuss the use of BI in Finance.
- Q15.** Justify BI is useful for Telecommunications services.
- Q16.** Write short note on Business Intelligence and Inventory Management System.
- Q17.** Comment on: How might you implement BI findings within an organization?
- Q18.** Explain Role of BI in ERP.
- Q19.** Write short note on BI Applications in Logistics.